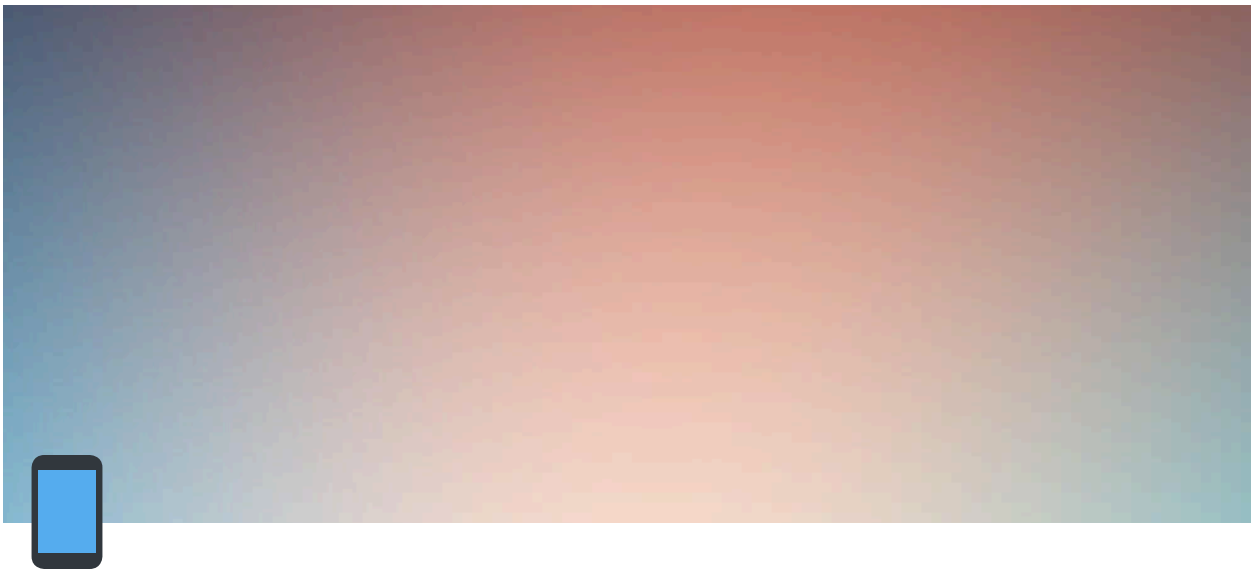

(Android) Create a branded app with full plug in Support

Problem

Currently, the client has a web application (Sprout - LMS) built on the Moodle framework. However, a significant portion of its users are students who find it inconvenient to always access the application on laptops or desktops. Moreover, mobile devices are more widely available to users compared to laptops, which aren't always accessible.

🎯 Goals

- To create an Android application that incorporates most of the features available on the web application
  - Third-party plugins added to the browser version should be supported, including interactive content (H5P) and other extensions.
  - All Moodle reporting features present in the customized browser version should be reflected in the app.

- Develop white-labeled features compatible with the mobile app
  - List of white-labeled features to be added after discussion

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- Ensure that key features are enhanced and seamlessly integrated into the application
  - Live classes through video conferencing tools (e.g., Zoom, Teams, Google Meet) (Final list of conferencing tools to be determined)
  - Content download feature for offline access
  - Compiler supporting multiple languages with enhanced performance
  - Notifications (Verify if notifications can be easily configured)

Assumption

> 💡 The Android application is being developed primarily for content consumption, specifically to be used by students

Constraints

- Feature Limitations: Not all Moodle framework features may be compatible with the Android platform.
- Third-Party Plugin Support: Some web-based plugins may not be directly supported in the mobile application.

📋 List of Plugins (Current integration in web application)

Assessment & Quiz Tools:

`CodeRunner`

`Realtime quiz`

`Quizventure`

Collaboration & Conferencing:

`BigBlueButton`

Interactive Learning Tools:

H5P

Game

Virtual programming lab

Storage & Integration:

Amazon S3

Source: Details as shared by Geekdino team in recent mail.

💡 Teams integration is needed within the app

Sitemap

https://miro.com/app/board/uXjVKi2wPO4=/?share_link_id=353939283390

Requirements

Requirement	User Story	Acceptance Criteria
Mobile friendly application	As a user, I want to access my LMS tool anytime and anywhere on my mobile device, so that I can manage my learning, interact with course content, and stay updated on assignments seamlessly, just like on the web application.	
Reporting Features	As a user, I want to view my learning analytics and reports to track my progress and identify areas for improvement.	
Live Class Participation	As a user, I want to join live classes through video conferencing tool (e.g., Zoom, Teams, Google Meet) to participate in real-time learning sessions.	

Requirement	User Story	Acceptance Criteria
Offline Access	As a user, I want to download course materials for offline access so that I can study without an internet connection.	
Compiler Functionality	As a user, I want to use a compiler within the app that supports multiple programming languages to help me complete coding assignments effectively.	
Notifications	As a user, I want to receive notifications about assignment deadlines, new course materials, and upcoming tests/exams so that I stay informed and organised.	

Core modules in the application

The core modules/menu options available in the web application currently are as follows:

Module/Menu options	Available in web application	Description	Required in Mobile App (First Release)
Home	✅	* This page contains the available courses card. * The view all button on this page directs to courses page	✅
Dashboard	✅	* Metrics * Timelines * Calendars	✅
My courses	✅	Contains list of enrolled courses	✅
Reports & Analytics	✅	Metrics & Graphs: * Engagement * Progress	✅ Only for teachers
Courses	✅	This page contains the available courses card with	Available in the home menu

Module/Menu options	Available in web application	Description	Required in Mobile App (First Release)
		different sort, filter & view options	
Notifications	✅	Toggle Notification Preferences: * Assignment Notification * Feedback Notification * Lesson Essay Graded Notification * System Notifications * Event Reminders	✅
Messaging	✅	Message with team and connect within the groups and can filter out few chats to private. Settings within the messaging tab will enhance privacy & Notification preferences.	❌

> The core modules are same for students & professors login/account. But the features or controls within each module/menu may be different for students & professors

User menu/options in the application (Student login)

Pages	Available in web application	Description	Required in Mobile app
Profile	✅	User profile section with edit option (3 clickable items)	✅
Grades	✅	Grades of the courses after evaluation	✅
Calendar	✅	Calendar tab to schedule events	✅
Private files	✅	Files tab to upload documents or notes	✅
Reports	✅	Reports based on the course performance	✅

Pages	Available in web application	Description	Required in Mobile app
Preferences	✅	An edit profile, change password and preferred language option based on user preference (15 Clickable items)*	✅
Logout	✅	To logout from user account	✅

Content Activities in the Application

Activities	Description	Available in Moodle Application	Modules used in the demo course (Sprout)
Assignments	Enable teachers to grade and give comments on uploaded files and assignments created on and off line	✅	✅
Big Blue Button	Run live video conferencing sessions within Moodle	✅	
Chat	Allows participants to have a real-time synchronous discussion	✅	
Choice	A teacher asks a question and specifies a choice of multiple responses	✅	
Database	Enables participants to create, maintain and search a bank of record entries	✅	✅
Feedback	For creating and conducting surveys to collect feedback.	✅	✅
Forum	Allows participants to have asynchronous discussions	✅	✅

Activities	Description	Available in Moodle Application	Modules used in the demo course (Sprout)
Glossary	Enables participants to create and maintain a list of definitions, like a dictionary	✅	
H5P Activity	Enables H5P content created in the Content bank or on h5p.com or with the lumi App to be easily added to a course as an activity.	✅	✅
Lesson	For delivering content in flexible ways	✅	✅
LTI External Tool Activity	Allows participants to interact with LTI compliant learning resources and activities on other web sites.	✅	✅
Quiz	Allows the teacher to design and set quiz tests, which may be automatically marked and feedback and/or to correct answers shown	✅	
SCORM	Enables SCORM packages to be included as course content	✅	
Survey	For gathering data from students to help teachers learn about their class and reflect on their own teaching	✅	
Wiki	A collection of web pages that anyone can add to or edit	✅	
Workshop	Enables peer assessment	✅	

UI/UX

Check if the Moodle framework allows us to customise the UI/UX for the Android application.

Out of scope

- Edit mode switch presented in the web application will not be present in the android application in order to keep the UI intact & user experience streamlined through the application.
- Light/Dark mode switch
- Chat module (not included for first release as of now)